

Stardog Designer with Voicebox Exercise Guide

This guide will provide information to help users complete class exercises and connect to the training environment for the Stardog Designer with Voicebox.

This exercise uses a customer 360 example. Users will be able to model, query, and explore information about a customer's purchase history and behavior, customer loyalty accounts, product categories and the most used payment methods to name a few. The exercises are designed to help users learn how to build a data model using .CSV files containing synthetic data.

This guide teaches you how to use Designer on endpoints with Voicebox enabled.

Please note: *The Cloud applications are constantly changing and though we try to keep this guide up to date, sometimes the screenshots and steps will vary. In this case, email training@stardog.com for assistance.*

Training Environment

Training Endpoint: Will be provided by your Instructor. If you are using an endpoint without Voicebox, you can use this [Designer Guide](#).

Data Modeling with Stardog Designer

This example shows how a user can build a data model from scratch using Stardog Designer. In this section, users will create a Designer project and then start building out a data model. There are multiple ways to do this, but in this section, a user will start by building the model in Designer.

The training goes over the process to design a data model. These exercises will go over how to build this in Designer. Here is an example data model for this use case. The models will vary depending on the competency questions that are used to help generate the model.

Example Designer View (with mappings)

*If your endpoint is not Voicebox enabled, you will only have to complete Step 1.

- Name your project. For example: **Store**
- Enter this **Description**
 - **The Customer 360 project creates a single, comprehensive view of each customer by integrating data from sales, marketing, service, and billing systems. With a unified customer view, the business can improve personalization, identify cross-sell opportunities, and strengthen customer retention strategies. Entities that should be highlighted are Customer, Purchase, Product, Category, and Store. Customers make purchases of products with and without rewards accounts. These purchases contain products that are in product categories.**
- **Competency Questions** - Review the auto generated Competency Questions. You can edit, add, change or delete any of them in the list.
- Add in the following questions:
 - How many customers shopped in Online stores vs Physical Stores?
- **Data Connection** - connect data. Here is the list of files to download. Once downloaded, connect all of these .CSV files.

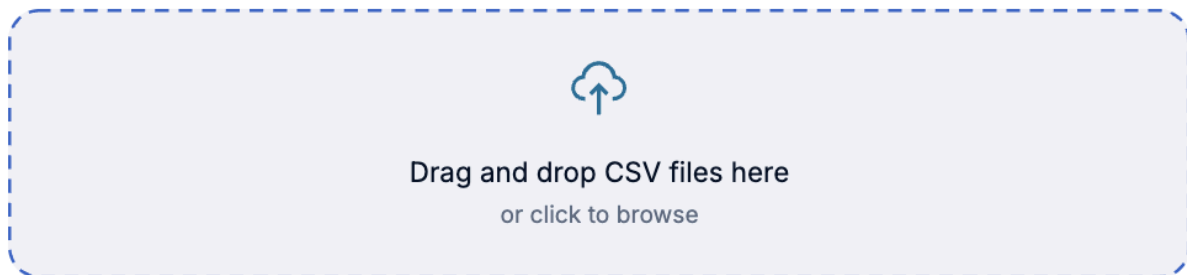
Download these files locally:

- [Customer](#)
- [Customer](#)
- [Product](#)
- [Purchase](#)

These are also available on Github:

- <https://github.com/stardog-union/training/tree/main/designer/v11.2>

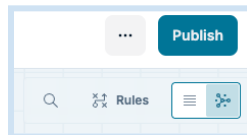
Drag and drop the files into this box and then click **Generate Model**.



Exercise 2 - View Project Settings

Step 1 - View Default Settings

- Click on the ... button on the top right of the canvas to access Settings.
- In v11.1+ the default namespace is: **urn:[Project Name]:**
- Layout: change the layout of the canvas
- AI mapping assist - this allows you to turn off the AI mapping assistance.



Project Settings

Project Name: Store

Project Description: The Customer 360 project creates a single, comprehensive view of each customer by integrating data from sales, marketing, service, and billing systems. With a unified customer view, the business can improve personalization, identify cross-sell opportunities, and strengthen customer retention strategies.

Layout: ☐ Force-directed ☒ Horizontal Tree ☐ Vertical Tree

AI mapping assist (Recommended): ☒

Instance Data Namespace: urn:Store:

Infer class types based on shared properties: ☐ Yes ☒ No ☐ Custom per property

Data Model	Namespace	Set Prefix	Prefix	AI Hints
Store	urn:Store:	☒	Store	☒

Buttons: Cancel, Save

Exercise 3 - Create Attributes

Step 1 - Review the Data Model

- Review the data model that was generated

Class Detail

Tabs: Basic | Hierarchy | Relationships | **Attributes** | Geospatial

Attribute Name	Data Type	Rule
Id	string	Create
Price	decimal	Create
Quantity	integer	Create
Time	string	Create

A search dropdown is open for 'Id', showing 'Rewards Account' and 'Create "Rewards Account"' as options.

Step 2 - Create a Rewards Account Attribute

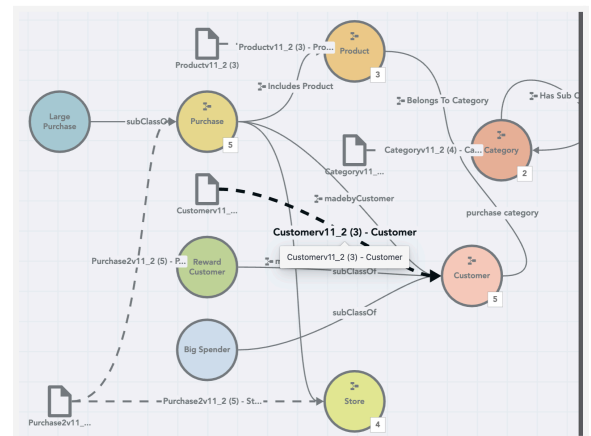
- Double click on the Purchase Class
- Name the attribute: **Rewards Account**
- Select **String** as the data type
- Click **Add**

Exercise 4 - Map the data

Voicebox suggests mapping during the project generation stage. It is a good best practice to review the primary Identifiers, labels and relationships. In this example, you will review a couple mappings.

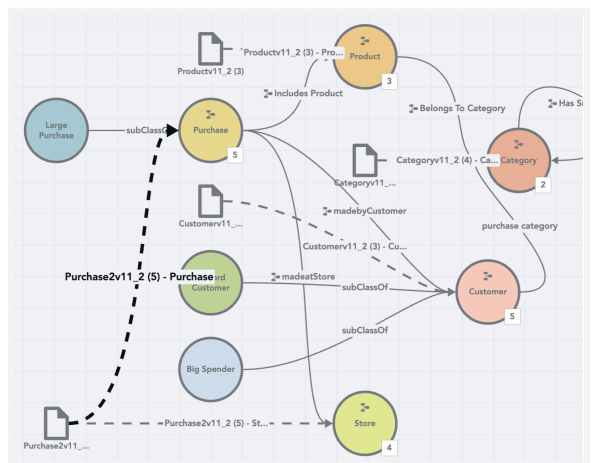
Step 1 - Review Customer

- Double click on dotted line between the Customer resource and the Customer Class
- Verify that the primary identifier is cid and the label is full_name. If it isn't, make the change.



Step 2 - Map the Rewards Account attribute

- Double click on the dotted line between the Purchase resource and the Purchase Class
- Under attributes, find the Rewards Account attribute. If this is blank, select the reward_account attribute from the drop-down list. This step maps the column named reward_account from the CSV file to the Rewards Account attribute in the Purchase Class.



Store Data Model Mapping Publish

Purchase2v11_2 (5) Type: CSV Suggest Mapping

Primary Identifier: Preview: bad78090-e88e-40f5-ad75-6329445a739e

Label:

Relationships

- Includes Product: Target Class: Product, Target Identifier: id, Preview: 176
- Made By Customer: Target Class: Customer, Target Identifier: cid, Preview: 206

Attributes

Attribute	Type	Value
Date	string	date
Price	decimal	price
Quantity	integer	quantity
Rewards Account	string	reward_account

Rewards Account string

Step 3 - Review the Store Class

- Review the Store Class to see which attributes were created. To add attributes, double click on the Store class and select Attributes. You will need the following attributes:
 - Store_address = string
 - Store_type = string
 - Store_lat = float
 - Store_long = float
- *Optional:* Use the list below to add additional attributes.

Attribute Name	Data Type	Rule
		Add
store_address	string	Create
store_address line 1	string	Create
store_address line 3	string	Create
store_city	string	Create
store_latitude	float	Create
store_longitude	float	Create
store_state	string	Create
store_type	string	Create
store_url	anyURI	Create
store_zip	string	Create
storeID	integer	Create

- **Tip:** If you need to edit an attribute in the future, you can also view a list by switching into **List view**. Click on the attribute and edit the data type.

Data Model Mapping			
Classes Relationships Attributes			
Attribute Name	Data Types	Associated Clas...	KGTraining...
parent	string	Categoryv1_2 (2)	●
phone	string	Customer	●
pid	integer	Purchasev1_2 (2)	●
price	string	Purchasev1_2 (2)	●
product	string	Product	●
quantity	integer	Purchasev1_2 (2)	●
reward_account	string	Purchasev1_2 (2)	●
ssn	string	Customer	●
Store_Address	string	Purchasev1_2 (...)	●
Store_City	string	Purchasev1_2 (...)	●
Store_State	string	Purchasev1_2 (...)	●
Store_Type	string	Purchasev1_2 (...)	●
Store_URL	anyURI	Purchasev1_2 (2)	●
Store_Zip	string	Purchasev1_2 (...)	●
StoreID	integer	Purchasev1_2 (...)	●
time	string	Purchasev1_2 (2)	●
vendor	string	Product	●

Step 4 - Review the Store Mapping

- Review the mapping suggestions. Mapping suggestions can vary based on the competency questions, description and data used for your project.
- Double click on the dotted line between the Purchase resource and the Store Class. Verify that your Primary identifier and label are correct.
- Map the new attributes: Under Attributes, select the correct column name from the drop-down to map to the new attributes.

Purchasev1_2 (2) - ... Store		Purchasev1_2 (2) Type: CSV		Suggest Mapping	
Primary Identifier ⓘ		StoreID x			
Preview: 1					
Label ⓘ		StoreID			
Relationships					
Attributes					
Store_Address	string	Store_Address		...	
Store_City	string	Store_City		...	
Store_State	string	Store_State		...	
Store_Type	string	Store_Type		...	
Store_URL	anyURI	Store_URL		...	
Store_Zip	string	Store_Zip		...	
StoreID	integer	StoreID		...	

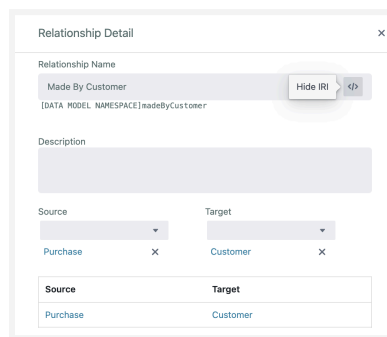
- Navigate back to the **Data Model** view.

Exercise 5 - Edit Model Objects

Classes, relationships and attributes can be edited from the UI. In this section, all relationships will be renamed so that they are in camel case. For example, Made by Customer becomes *madebyCustomer*.

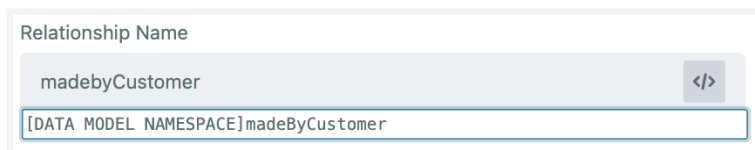
Step 1 - Edit Relationship

- Double-click on the relationship between Purchase and Customer.
- In the Relationship Detail, change the name to madebyCustomer.



The screenshot shows a 'Relationship Detail' window. At the top, the 'Relationship Name' is 'Made By Customer' with a 'Hide IRI' button and a code icon. Below it, the IRI is '[DATA MODEL NAMESPACE]madeByCustomer'. There is a 'Description' field. Under 'Source' and 'Target', there are dropdown menus showing 'Purchase' and 'Customer' respectively, with an 'X' icon next to each. At the bottom, there is a table with two columns: 'Source' and 'Target', with 'Purchase' and 'Customer' listed below them.

- Click on the Code Block icon  to show IRI. Edit IRI to be **[MODEL NAMESPACE]madebyCustomer**.



The screenshot shows the 'Relationship Detail' window after editing. The 'Relationship Name' is now 'madebyCustomer'. The IRI field is highlighted and contains '[DATA MODEL NAMESPACE]madeByCustomer'. The 'Source' and 'Target' dropdowns still show 'Purchase' and 'Customer'.

- Close the Relationship Detail window with the **X** button in the upper right-hand corner.
- Repeat these steps for all of the relationships.

Create & Map Relationships

These exercises will walk you through validating relationships. Relationships describe how two classes are connected. For example, a Customer made purchase describes the relationship between Customer and Purchase.

Exercise 6 - Edit Map Relationships

Step 1 - Edit the mappings

Edit the mappings by making sure each primary key has been identified and the relationship target and source have identifiers. Users can also update Labels too.

Label annotations are used throughout Stardog as the display name for an instance. When no label is set, the primary identifier value will be displayed instead.

- Double-click on the **Purchase** file in the canvas.
- Click on the mapping for **Purchase - Purchase**.
- Verify that the **Label** mapping is `id`.

***The relationship names will vary from student to student.**

- Set the **Target Class** for the relationship **for product** to **Product** and the **Target Identifier** to **pid**.
- Set the **Source Class** for the relationship **made purchase** to **Customer** and the **Source Identifier** to **cid**.
- Click on Data **Model** to go back to the canvas.

Relationships		
Includes Product	Target Class Product	Target Identifier pid Preview: 176
madebyCustomer	Target Class Customer	Target Identifier cid Preview: 206

Publish a Data Model

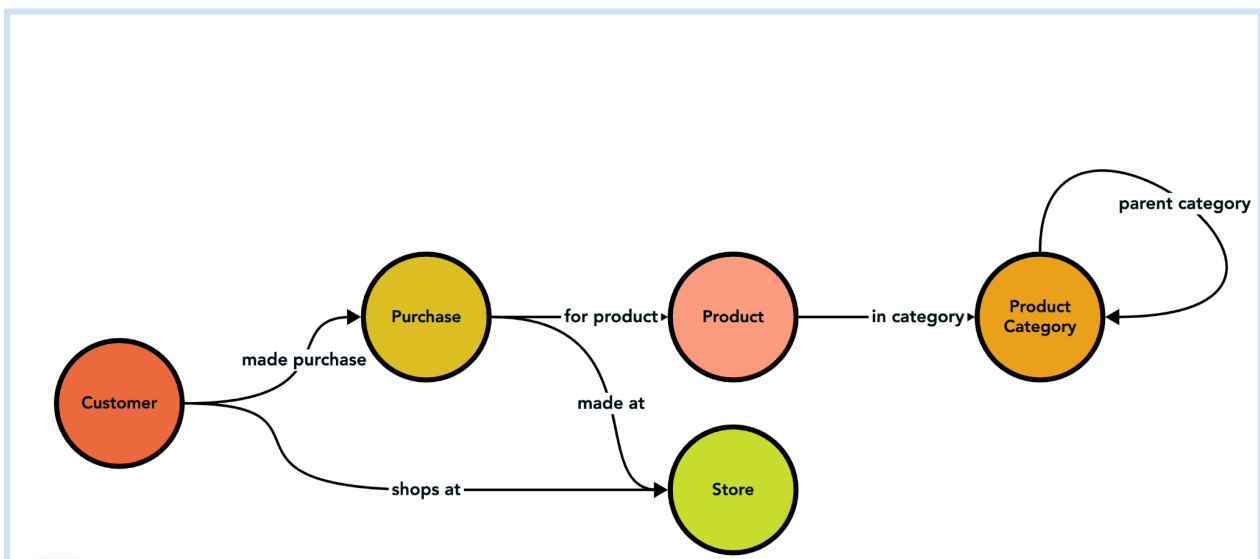
In this step, you will publish your project so that you can view it in Stardog Explorer. This allows you to see the model and view the mapped data. This is a good way to validate your work.

Exercise 7 - Publish your project

To publish you will need to:

- Click on Publish in the top right corner
- Find the database - `[Training\_Your initials]`

- a. Make sure that Enable Voicebox and Publish questions as spotlight questions are enabled.
- Accept the default model namespace or modify - Accept
 - By default, the “Download zipped project is turned on. Toggle off if you don’t want to have a .zip file downloaded each time you publish your project.
 - Click Next
 - You can adjust the model configuration. For this exercise just accept the defaults which will create a new model.
 - Click Next
 - You can adjust the mappings. For this exercise just accept the defaults, which will create new graphs for all resources.
 - Click Publish
 - Click on the Stardog Explorer image or navigate there by going to cloud.stardog.com, Select Explorer, connect to your database, and then click on Visualize.



*Your model might be slightly different based on the inputs used to create the model .

Inference Rules

This section will cover the steps to create new classes and inference rules. These will then be published to Explorer. Inference rules infer new data from explicit data. Inference rules can be set up using classes or relationships.

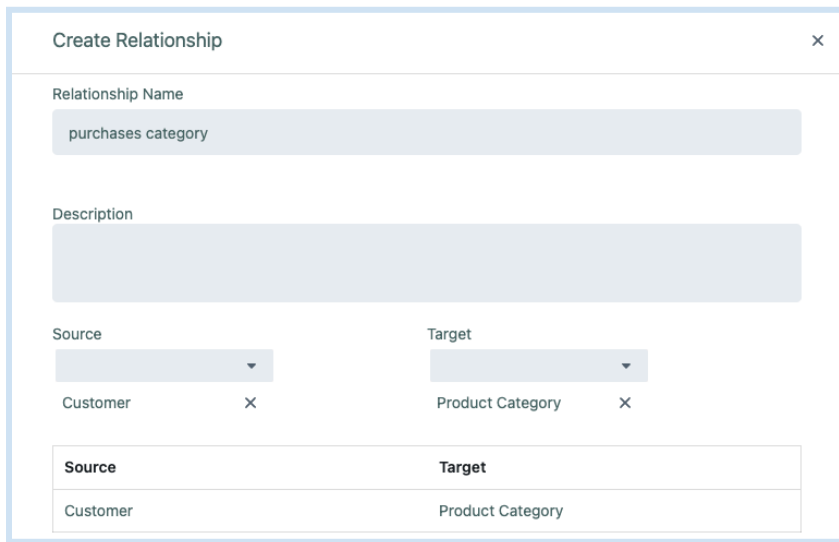
Exercise 8 - Create Inference Rules

Step 1 - Create a new relationship

From the Designer home screen, click on the **Add Relationship** button on the tool bar.



Create a new relationship for **purchases category** selecting **Customer** as the **Source** and **Product Category** as the **Target**.

A screenshot of the 'Create Relationship' dialog box in Stardog. The dialog has a title bar 'Create Relationship' with a close button 'x'. It contains three main sections: 'Relationship Name' with a text input field containing 'purchases category'; 'Description' with a large empty text area; and 'Source' and 'Target' dropdown menus. Below the dropdowns, 'Customer' is selected for Source and 'Product Category' is selected for Target, each with a small 'x' icon to its right. At the bottom, there is a table summarizing the selections.

Source	Target
Customer	Product Category

Step 2 - Create a rule for the new relationship

- Click on the Rules button  **Rules**
- Select **Create Rule**
- Name the Rule: **Purchased Category**
- Add in the conditions listed below and then click save

Rule Detail

Rule Name

Purchased Category

Rule Builder

Text Editor

When the following conditions are met:

+

Customer

made purchase

→

Purchase

×

+

Purchase

for product

→

Product

+

Product

in category

→

Product Category

+

AND NOT

Then infer:

+

Customer

purchases category

→

Product Category

×

+

Cancel

Save

Please Note: It is a good best practice to publish the model frequently during development to validate changes by viewing it in Explorer.

Step 3 - Create rules referenced below

- Create the following additional rules:

Create Rule
Rule Name: Loyalty Customers Conditions: Customer - made purchase → Purchase Purchase - rewards_account exists Then infer: Customer Classify As Rewards Customer

Rule Name: Large Purchases

Conditions:

Purchase - price $\geq$ 20000

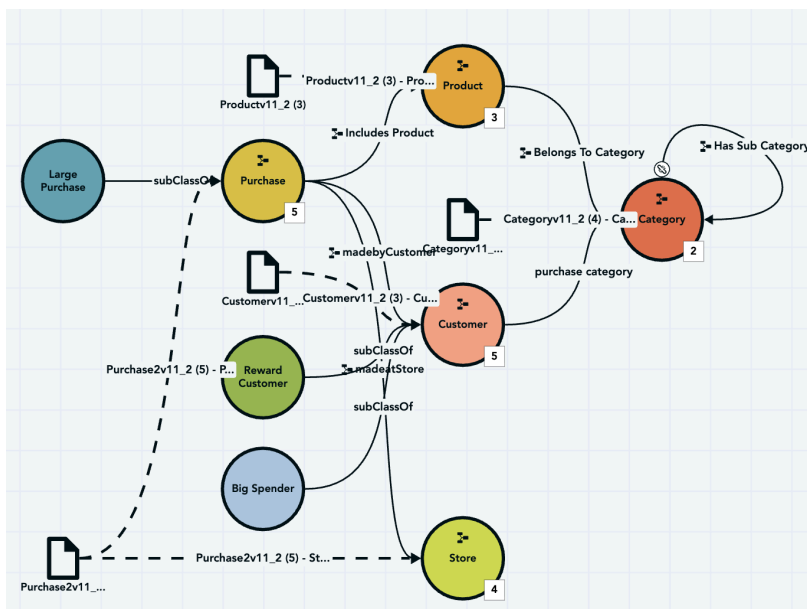
Then infer: Purchase classify As Large Purchase

Rule Name: Big Spenders

Conditions:

Customer - made purchase $\rightarrow$ Large Purchase

Then infer: Customer Classify As Big Spender



Exercise 9 - Publish your project

In this step, you will publish your project so that you can view it in Stardog Explorer. This allows you to see the model and view the data that was mapped. This is a good way to validate your work.

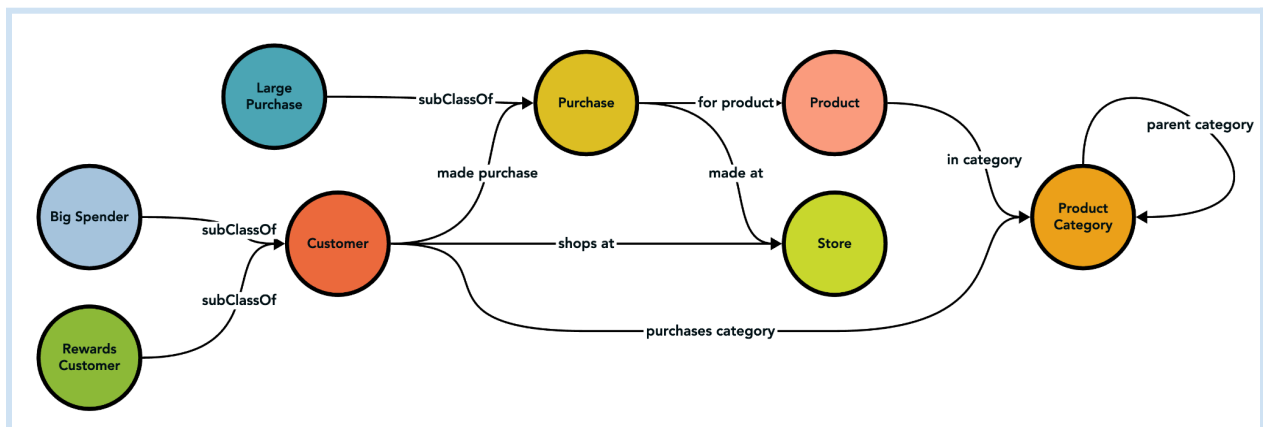
To publish you will need to:

- Select the Database you originally published to `Training\_[Your initials]`
- Accept the default model namespace or modify it.
- By default, the “Download zipped project when publishing” is turned on.
- Click Next

- You can adjust the model configuration. For this exercise just accept the defaults, which will create a new model. Make sure that the **Publish Inference Rules** is turned on.
- Click Publish
- Click on the link to open in Explorer.

Exercise 10 - View in Explorer

Now it is time to validate your inference rules by viewing your new published data model in Explorer. After publishing the new model, open your database in Explorer. Open Settings, and turn on Reasoning. The canvas should now display the data model with the new classes that were created to hold the inferred information.



Exercise 11 - Replace a Project Resource

Step 1 - Replace the Purchase project resource

This exercise demonstrates how to update a CSV file when its content changes, provided the column structure remains the same.

- Download and update Category .CSV
 - [Category_extended](#)
- Open a preview of the **Category** resource by clicking on the eye icon



- Select **Replace** and select the Category_extended2.csv.

- Select **Save**.

Exercise 12 - Map Geolocation data

Step 1 - Add new Geolocation attributes and map

- Double click the **Store** class.
- Go to **Attributes** and make sure that store latitude and store longitude has the **data type of float**.
- Go to **Geospatial** and set the latitude and longitude and then use the store address for some of the other map details.

Map Detail	Content
Latitude	store_latitude ▼
Longitude	store_longitude ▼
Node Size	▼
Popup Title	store_address lin... ▼
Popup Content	store_address lin... ▼
Tooltip Title	store_address lin... ▼
Tooltip Content	store_address lin... ▼

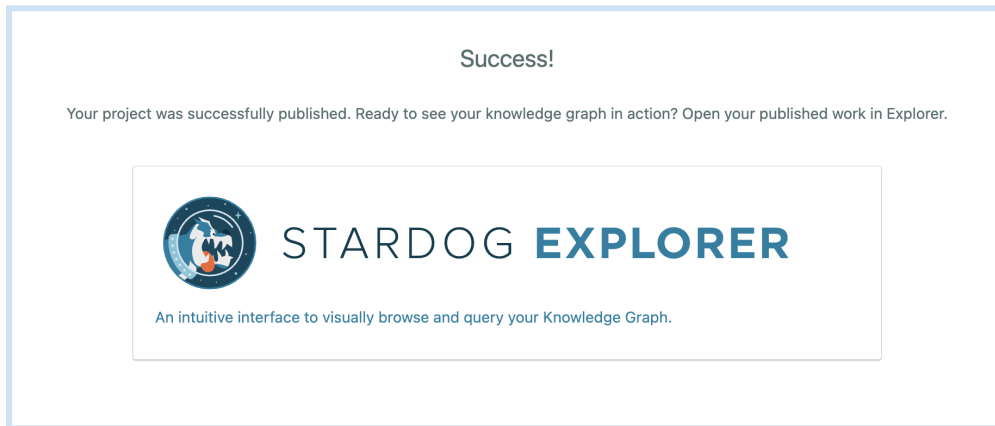
Step 2 - Publish

To publish you will need to:

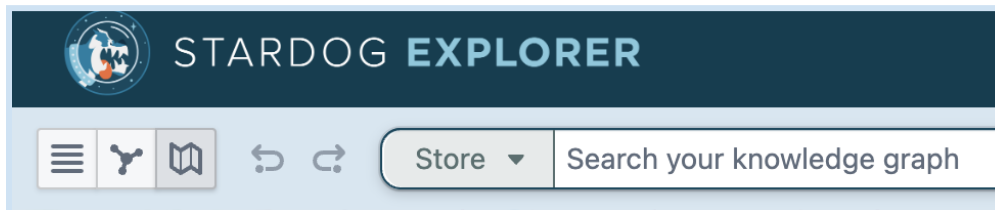
- Select the Database you originally published to - **Training_[Your initials]**
- Accept the default model namespace or modify it.
- By default, the **Download zipped project when publishing** is turned on. Turn it off.
- Click Next
- You can adjust the model configuration. For this exercise just accept the defaults, which will create a new model. Make sure that the **Publish Inference Rules** is turned on.
- Click **Publish**

Step 3 - View in Explorer

- Click **Stardog Explorer**



- Select **Store** in the class drop down and search



- Select the **Map** view

